

Quantium Virtual Internship - Retail Strategy and Analytics - Task

1

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Load libraries and data

```
library(dplyr)
library(ggplot2)
library(kableExtra)
```

```
chips <- read.csv('/Users/thibautraiola/DATA-ANALYST/Project4Portfolio/Forage/Quantium - DA /1. Data pr
```

Data overview

```
head(chips)
```

```
##          DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR
## 1 2018-10-17         1         1000      1         5
## 2 2019-05-14         1         1307    348        66
## 3 2019-05-20         1         1343    383        61
## 4 2018-08-17         2         2373    974        69
## 5 2018-08-18         2         2426   1038       108
## 6 2019-05-16         4         4149   3333        16
##
##          PROD_NAME PROD_QTY TOT_SALES PACK_SIZE
## 1  Natural Chip      Compny SeaSalt175g      2      6.0      175
## 2              CCs Nacho Cheese    175g      3      6.3      175
## 3  Smiths Crinkle Cut  Chips Chicken 170g      2      2.9      170
## 4  Smiths Chip Thinly S/Cream&Onion 175g      5     15.0      175
## 5 Kettle Tortilla ChpsHny&Jlpno Chili 150g      3     13.8      150
## 6 Smiths Crinkle Chips Salt & Vinegar 330g      1      5.7      330
##  BRAND_NAME      LIFESTAGE PREMIUM_CUSTOMER
## 1  Natural  YOUNG SINGLES/COUPLES      Premium
## 2          CCs MIDAGE SINGLES/COUPLES      Budget
## 3  Smiths MIDAGE SINGLES/COUPLES      Budget
## 4  Smiths MIDAGE SINGLES/COUPLES      Budget
## 5  Kettle MIDAGE SINGLES/COUPLES      Budget
## 6  Smiths MIDAGE SINGLES/COUPLES      Budget
```

```
summary(chips)
```

```
##      DATE      STORE_NBR  LYLTY_CARD_NBR      TXN_ID
## Length:248229   Min.    : 1    Min.    : 1000   Min.    : 1
## Class :character 1st Qu.: 70    1st Qu.: 70015   1st Qu.: 67564
## Mode  :character Median :130    Median : 130360   Median : 135150
##              Mean  :135    Mean  : 135524   Mean  : 135126
##              3rd Qu.:203    3rd Qu.: 203081   3rd Qu.: 202644
##              Max.   :272    Max.   :2373711   Max.   :2415841
##      PROD_NBR   PROD_NAME      PROD_QTY      TOT_SALES
## Min.    : 1    Length:248229   Min.    :1.000   Min.    : 1.700
## 1st Qu.: 27    Class :character 1st Qu.:2.000   1st Qu.: 5.800
## Median : 52    Mode  :character Median :2.000   Median : 7.400
## Mean    : 56              Mean  :1.906   Mean  : 7.303
## 3rd Qu.: 86              3rd Qu.:2.000   3rd Qu.: 8.800
## Max.    :114              Max.   :5.000   Max.   :29.500
##      PACK_SIZE   BRAND_NAME      LIFESTAGE      PREMIUM_CUSTOMER
## Min.    : 70.0   Length:248229   Length:248229   Length:248229
## 1st Qu.:150.0   Class :character  Class :character  Class :character
## Median :170.0   Mode  :character  Mode  :character  Mode  :character
## Mean    :175.4
## 3rd Qu.:175.0
## Max.    :380.0
```

```
str(chips)
```

```
## 'data.frame': 248229 obs. of 12 variables:
## $ DATE : chr "2018-10-17" "2019-05-14" "2019-05-20" "2018-08-17" ...
## $ STORE_NBR : int 1 1 1 2 2 4 4 5 7 7 ...
## $ LYLTY_CARD_NBR : int 1000 1307 1343 2373 2426 4149 4196 5026 7150 7215 ...
## $ TXN_ID : int 1 348 383 974 1038 3333 3539 4525 6900 7176 ...
## $ PROD_NBR : int 5 66 61 69 108 16 24 42 52 16 ...
## $ PROD_NAME : chr "Natural Chip Compny SeaSalt175g" "CCs Nacho Cheese 175g" "Smiths"
## $ PROD_QTY : int 2 3 2 5 3 1 1 1 2 1 ...
## $ TOT_SALES : num 6 6.3 2.9 15 13.8 5.7 3.6 3.9 7.2 5.7 ...
## $ PACK_SIZE : int 175 175 170 175 150 330 210 150 210 330 ...
## $ BRAND_NAME : chr "Natural" "CCs" "Smiths" "Smiths" ...
## $ LIFESTAGE : chr "YOUNG SINGLES/COUPLES" "MIDAGE SINGLES/COUPLES" "MIDAGE SINGLES/COUPLES"
## $ PREMIUM_CUSTOMER: chr "Premium" "Budget" "Budget" "Budget" ...
```

Data preparation

```
chips$DATE <- as.Date(chips$DATE, "%Y-%m-%d")
```

Helper functions

```
create_bar_plot <- function(data, x_col, y_col, title, x_label, y_label, fill_label = x_label, text_size = 12, data |>
```

```

ggplot(aes(x = reorder({{ x_col }}, -{{ y_col }}), y = {{ y_col }}, fill = {{ x_col }})) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = text_size)) +
  ggtitle(title) +
  labs(x = x_label, y = y_label, fill = fill_label)
}

```

Customer segments analysis

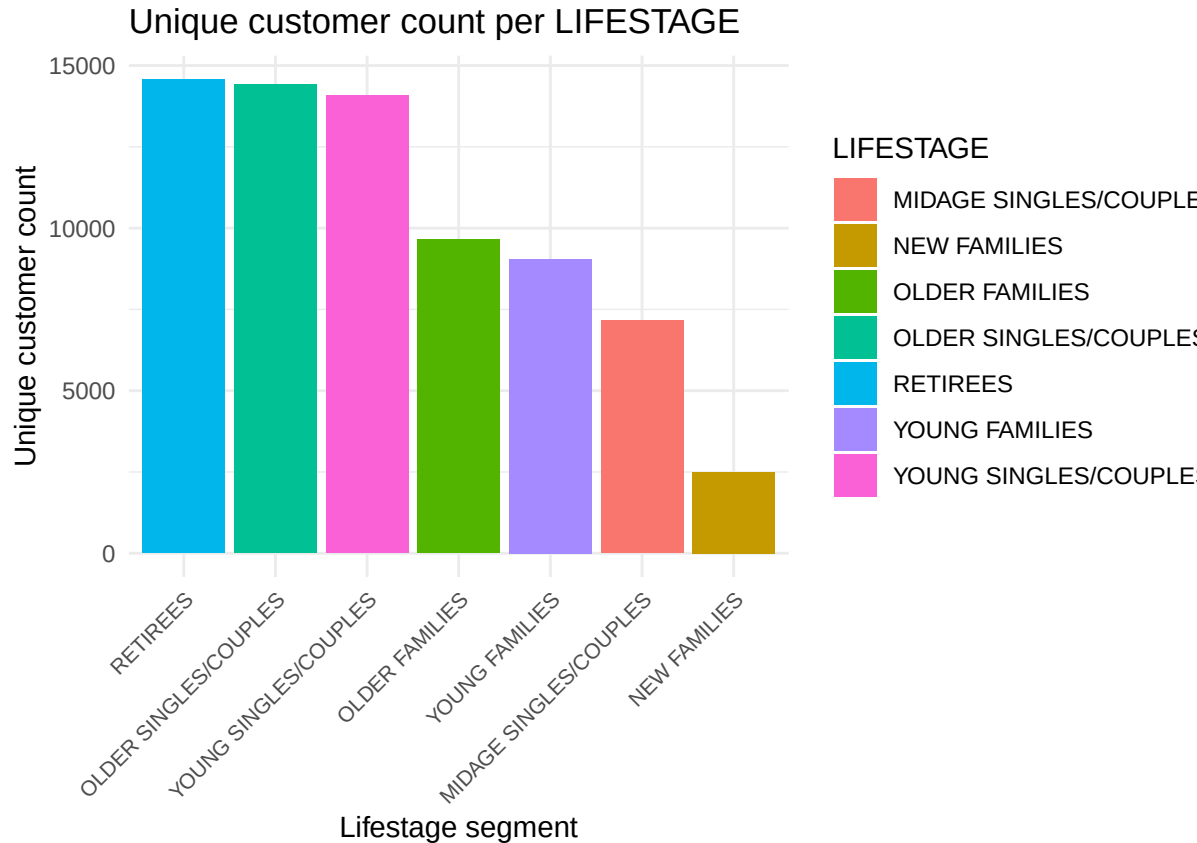
Number of unique customers per segment

```

customer_by_lifestage <- chips |>
  group_by(LIFESTAGE) |>
  summarise(n_customers = n_distinct(LYLTY_CARD_NBR)) |>
  arrange(desc(n_customers))

customer_by_lifestage |>
  ggplot(aes(x = reorder(LIFESTAGE, -n_customers), y = n_customers, fill = LIFESTAGE)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8)) +
  ggtitle("Unique customer count per LIFESTAGE") +
  labs(x = "Lifestage segment", y = "Unique customer count")

```

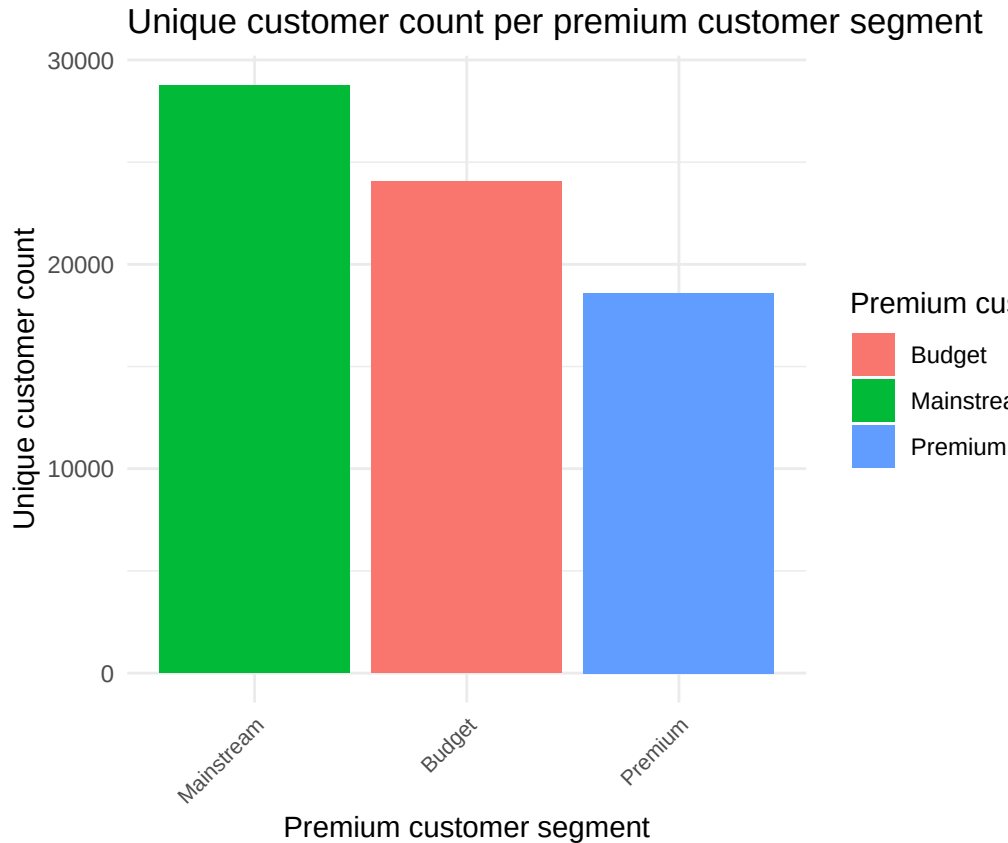


By LIFESTAGE

Retirees, older singles/couples, and young singles/couples segments dominate by far, each with more than 14k unique customers. Older and young families fall between 9k and 10k, midage singles/couples account for around 7k, and new families represent only about 2.5k.

```
customer_by_premium_customer <- chips |>
  group_by(PREMIUM_CUSTOMER) |>
  summarise(n_customers = n_distinct(LYLTY_CARD_NBR)) |>
  arrange(desc(n_customers))

create_bar_plot(
  data = customer_by_premium_customer,
  x_col = PREMIUM_CUSTOMER,
  y_col = n_customers,
  title = "Unique customer count per premium customer segment",
  x_label = "Premium customer segment",
  y_label = "Unique customer count"
)
```



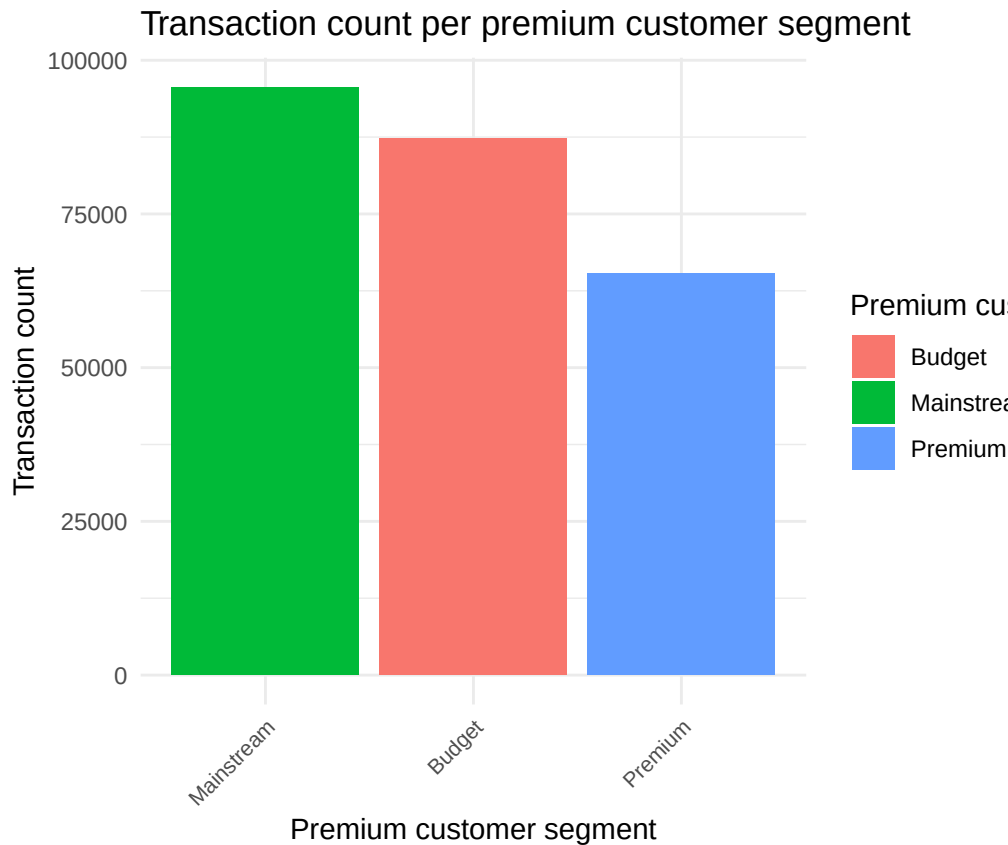
By PREMIUM_CUSTOMER

The mainstream segment naturally has the highest number of unique customers (29k), followed by budget (24k) and premium (19k).

Number of transactions per segment

```
transaction_by_premium_customer <- chips |>
  group_by(PREMIUM_CUSTOMER) |>
  summarise(n_transactions = n())

create_bar_plot(
  data = transaction_by_premium_customer,
  x_col = PREMIUM_CUSTOMER,
  y_col = n_transactions,
  title = "Transaction count per premium customer segment",
  x_label = "Premium customer segment",
  y_label = "Transaction count"
)
```

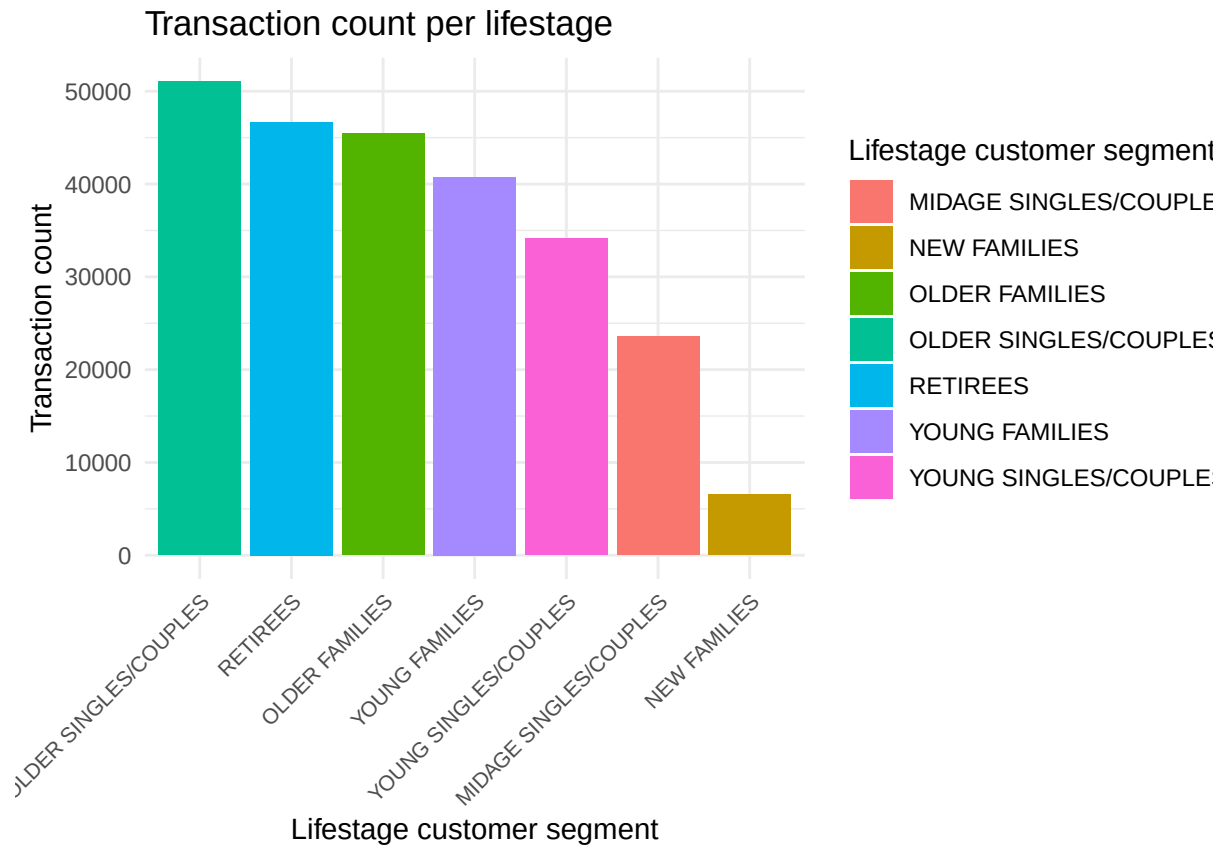


By PREMIUM_CUSTOMER

We observe that the mainstream segment naturally accounts for the highest number of transactions, around 94k, while the premium segment records just over 60k transactions, and budget sits around 87k.

```
transaction_by_lifestage <- chips |>
  group_by(LIFESTAGE) |>
  summarise(n_transactions = n())

create_bar_plot(
  data = transaction_by_lifestage,
  x_col = LIFESTAGE,
  y_col = n_transactions,
  title = "Transaction count per lifestage",
  x_label = "Lifestage customer segment",
  y_label = "Transaction count"
)
```



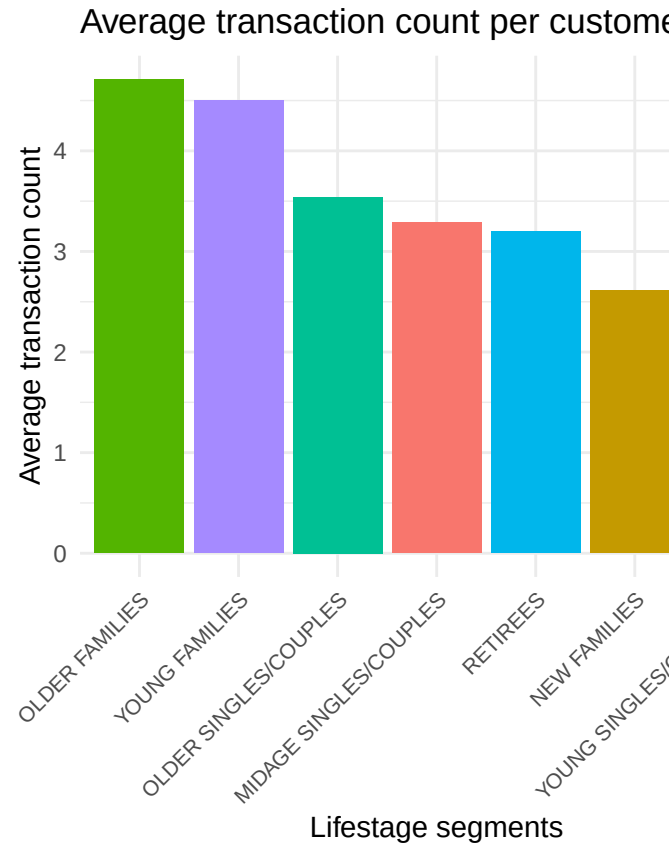
By LIFESTAGE

Here, the dominant segments are older singles/couples (50k), retirees (46k), and older families (45k), which is not surprising given that the first two also have the highest number of customers. However, while the young singles/couples group is the 3rd largest segment in terms of unique customers, it ranks only 5th in terms of transaction count. A further analysis of the average number of purchases per customer by segment will be necessary. Finally, the new families segment is by far the smallest, with barely 7k transactions, consistent with it also having the fewest unique customers.

```
transaction_per_customer <- chips |>
  group_by(LYLTY_CARD_NBR, PREMIUM_CUSTOMER, LIFESTAGE) |>
  summarise(n_transactions = n())

avg_transactions_by_lifestage <- transaction_per_customer |>
  group_by(LIFESTAGE) |>
  summarise(avg_transactions = round(mean(n_transactions), 2))

create_bar_plot(
  data = avg_transactions_by_lifestage,
  x_col = LIFESTAGE,
  y_col = avg_transactions,
  title = "Average transaction count per customer, by LIFESTAGE",
  x_label = "Lifestage segments",
  y_label = "Average transaction count"
)
```



Average transaction count per customer, by LIFESTAGE

The young singles/couples segment ranks last, with fewer than 2.5 transactions per customer. In contrast, the older families segment averages more than 4.5 transactions per customer, which explains the gap observed earlier. Additionally, the young families segment shows a strong signal, averaging 4.5 transactions per customer, well ahead of the 3rd-ranked group (3.5).

Financial analysis

Total sales by segment

```
sales_by_segment <- chips |>
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) |>
  summarise(total_sales = sum(TOT_SALES))

sales_by_segment |>
  arrange(desc(total_sales))
```

```
## # A tibble: 21 x 3
## # Groups:   LIFESTAGE [7]
##   LIFESTAGE      PREMIUM_CUSTOMER total_sales
##   <chr>          <chr>          <dbl>
## 1 OLDER FAMILIES Budget          157613.
## 2 YOUNG SINGLES/COUPLES Mainstream      147999.
## 3 RETIREES       Mainstream      145837.
## 4 YOUNG FAMILIES Budget          130352.
```

```
## 5 OLDER SINGLES/COUPLES Budget 128130.
## 6 OLDER SINGLES/COUPLES Mainstream 125178.
## 7 OLDER SINGLES/COUPLES Premium 124067.
## 8 RETIREES Budget 106276.
## 9 OLDER FAMILIES Mainstream 96927.
## 10 RETIREES Premium 91669.
## # i 11 more rows
```

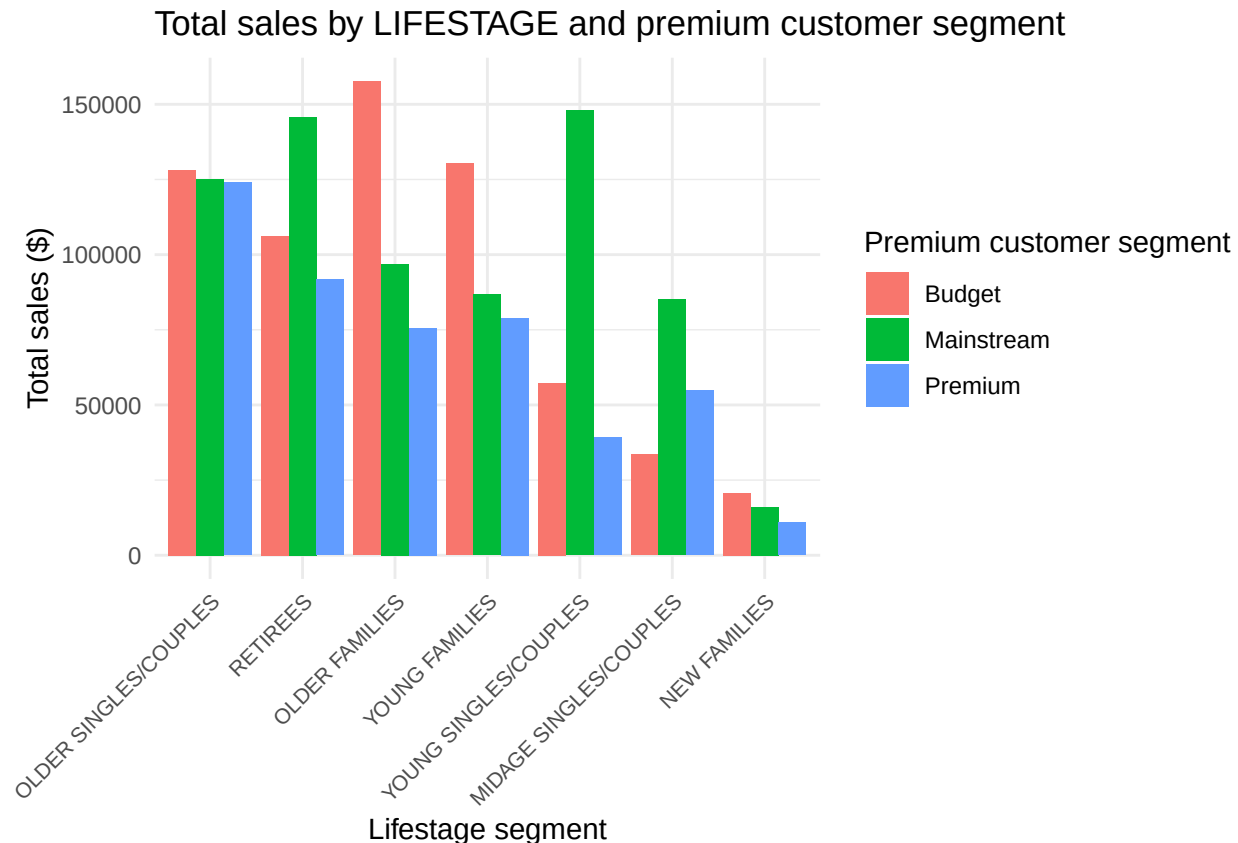
```
sales_by_lifestage_total <- sales_by_segment |>
  group_by(LIFESTAGE) |>
  summarise(total_sales = sum(total_sales)) |>
  arrange(desc(total_sales))

sales_by_lifestage_total |>
  kable(caption = "Total sales by LIFESTAGE") |>
  kable_minimal()
```

Table 1: Total sales by LIFESTAGE

LIFESTAGE	total_sales
OLDER SINGLES/COUPLES	377375.15
RETIREES	343781.90
OLDER FAMILIES	330180.00
YOUNG FAMILIES	296015.40
YOUNG SINGLES/COUPLES	244478.10
MIDAGE SINGLES/COUPLES	173391.70
NEW FAMILIES	47510.15

```
sales_by_segment |>
  ggplot(aes(x = reorder(LIFESTAGE, -total_sales, sum), y = total_sales, fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = "dodge") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8)) +
  ggtitle("Total sales by LIFESTAGE and premium customer segment") +
  labs(x = "Lifestage segment", y = "Total sales ($)", fill = "Premium customer segment")
```



First, no LIFESTAGE segment has the Premium customer segment as its main contributor to sales. Only the ‘Older singles/couples’ segment shows notable interest from Premium customers, with an almost even split across the 3 customer segments, and the ‘Midage singles/couples’ LIFESTAGE, where Premium customers rank 2nd, well ahead of Budget customers.

The Mainstream customer segment is particularly dominant among ‘Young singles/couples’, reflecting the mainstream positioning of these products. In fact, for this specific case, the Mainstream segment outweighs the other two segments combined, a unique case here. Among ‘Retirees’, the order is the same (Mainstream - Budget - Premium), but with a strong, though less overwhelming, dominance of the first.

Regarding the Budget customer segment, a fairly similar situation emerges: among ‘Older families’ and ‘Young families’, this segment clearly dominates sales; the order remains the same, with Mainstream in 2nd position, closely followed by Premium.

Finally, ‘New families’ show a total sales figure so low that it risks skewing the analysis, but contributions are fairly balanced across segments: Budget represents twice as much as Premium, with Mainstream sitting in between.

Average price per unit, by LIFESTAGE

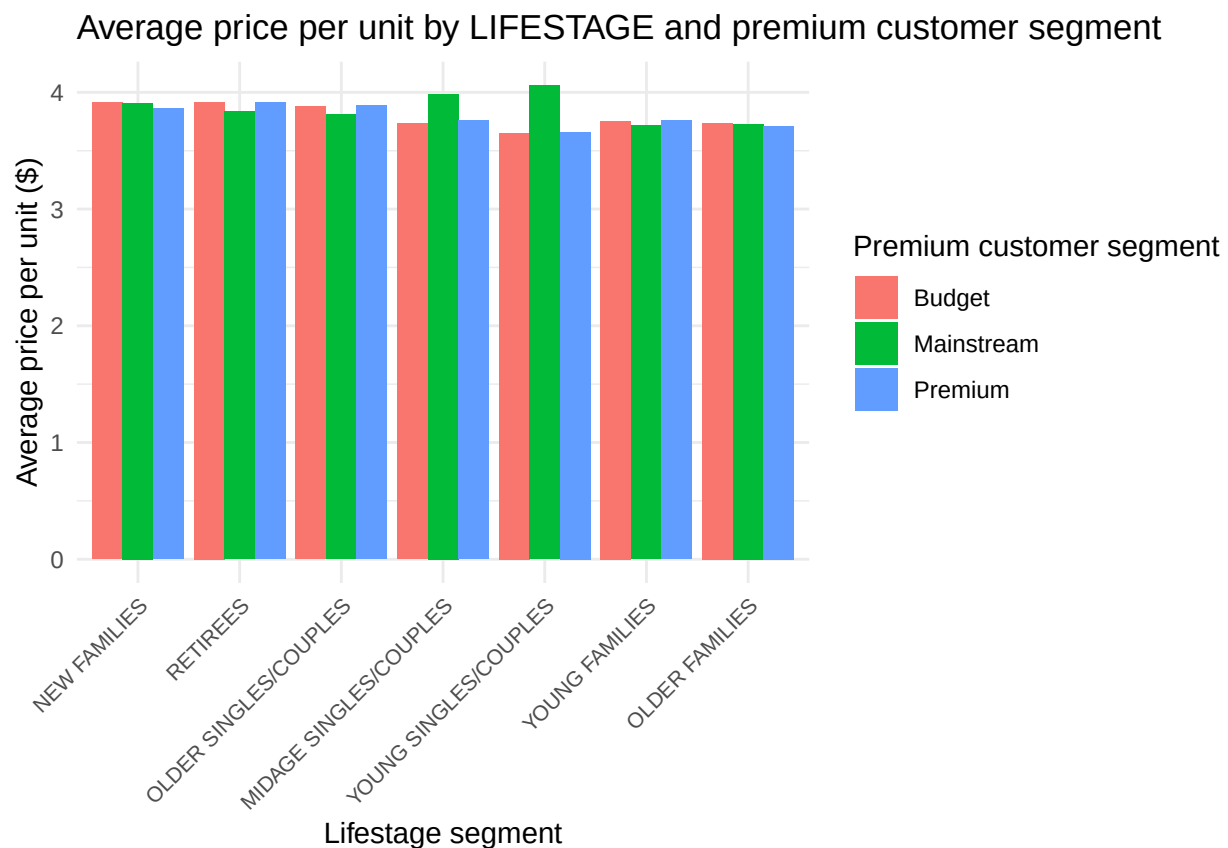
```
chips <- chips |>
  mutate(price_per_unit = TOT_SALES / PROD_QTY)
```

```
avg_price_per_unit <- chips |>
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) |>
  summarise(avg_price_per_unit = mean(price_per_unit))
```

```
avg_price_per_unit |>
  group_by(LIFESTAGE) |>
  summarise(avg_price = round(mean(avg_price_per_unit), 2)) |>
  arrange(desc(avg_price))
```

```
## # A tibble: 7 x 2
##   LIFESTAGE      avg_price
##   <chr>         <dbl>
## 1 NEW FAMILIES      3.89
## 2 RETIREES          3.89
## 3 OLDER SINGLES/COUPLES 3.86
## 4 MIDAGE SINGLES/COUPLES 3.83
## 5 YOUNG SINGLES/COUPLES 3.79
## 6 YOUNG FAMILIES     3.74
## 7 OLDER FAMILIES     3.73
```

```
avg_price_per_unit |>
  ggplot(aes(x = reorder(LIFESTAGE, -avg_price_per_unit), y = avg_price_per_unit, fill = PREMIUM_CUSTOMER_SEGMENT)) +
  geom_bar(stat = "identity", position = "dodge") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8)) +
  ggtitle("Average price per unit by LIFESTAGE and premium customer segment") +
  labs(x = "Lifestage segment", y = "Average price per unit ($)", fill = "Premium customer segment")
```



Overall, the differences in average price per unit are not significant across the different LIFESTAGE segments, all ranging between \$3.73 and \$3.89. Moreover, with rare exceptions, within a given LIFESTAGE segment, the different customer segments show relatively balanced averages.

The LIFESTAGE segments with the highest average price per unit are new families and retirees, both averaging \$3.89 across all customer segments combined. At the other end, older families and young families show the lowest averages, at \$3.73 and \$3.74 respectively.

However, among midage singles/couples and young singles/couples, the Mainstream segment shows a notably higher average compared to the other customer segments within the same group. In fact, only the Mainstream segment among young singles/couples has an average above \$4.

Statistical significance test

```
target_segment <- chips |>
  filter(LIFESTAGE %in% c("YOUNG SINGLES/COUPLES", "MIDAGE SINGLES/COUPLES"))

mainstream_prices <- target_segment |>
  filter(PREMIUM_CUSTOMER == 'Mainstream') |>
  pull(price_per_unit)

budget_premium_prices <- target_segment |>
  filter(PREMIUM_CUSTOMER != 'Mainstream') |>
  pull(price_per_unit)

t.test(mainstream_prices, budget_premium_prices)
```

```
##
##  Welch Two Sample t-test
##
## data:  mainstream_prices and budget_premium_prices
## t = 37.796, df = 55213, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.3166391 0.3512758
## sample estimates:
## mean of x mean of y
##  4.033359  3.699402
```

With a p-value below 2.2e-16, the observed difference between customer segments is very likely real. The \$0.33 difference, or 9%, indicates a non-negligible gap at scale. Finally, we can be 95% confident that the true difference between the observed segments lies between \$0.316 and \$0.351.

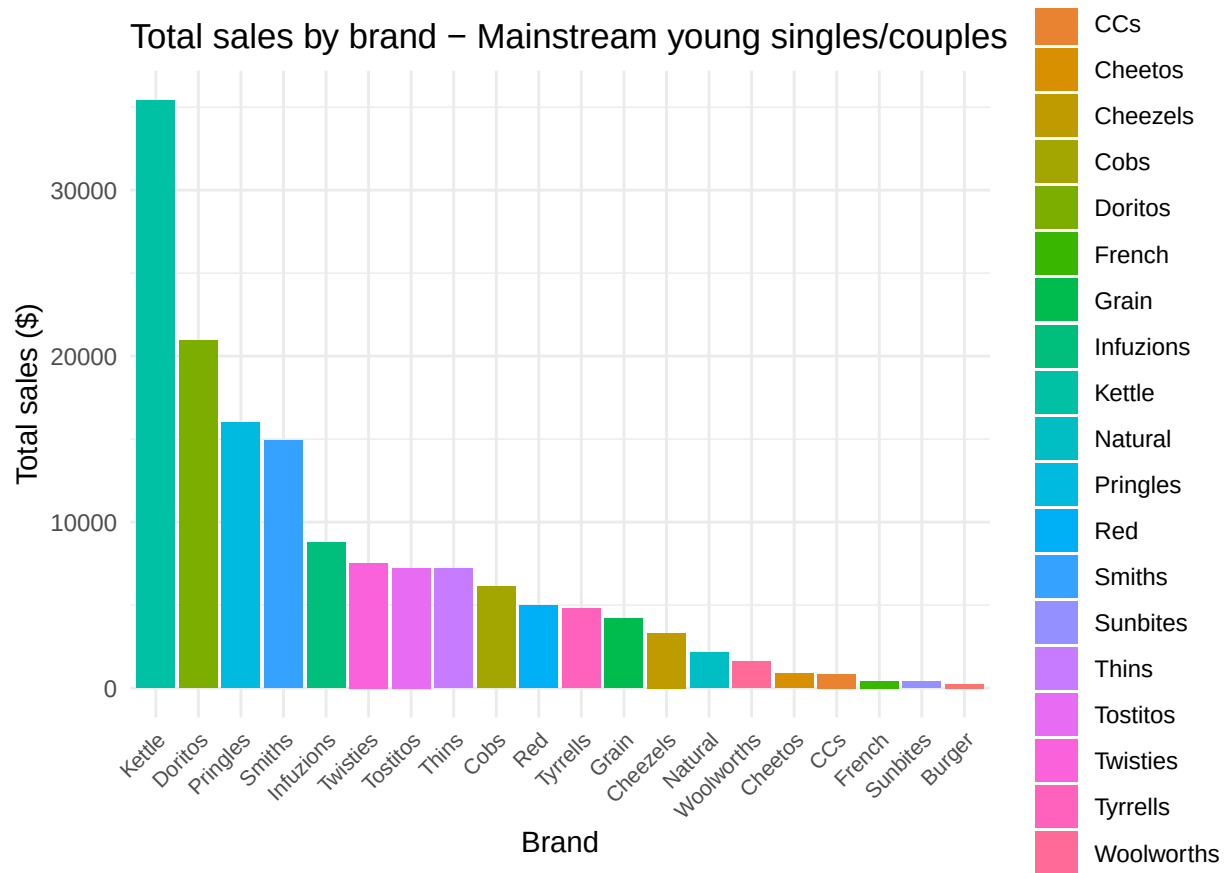
Deep dive - Mainstream young singles/couples

```
revenue_by_brand_target_segment <- chips |>
  filter(PREMIUM_CUSTOMER == 'Mainstream') |>
  filter(LIFESTAGE == 'YOUNG SINGLES/COUPLES') |>
  group_by(BRAND_NAME) |>
  summarise(tot = sum(TOT_SALES))
```

```
revenue_by_brand_target_segment |>
  arrange(desc(tot))
```

```
## # A tibble: 20 x 2
##   BRAND_NAME    tot
##   <chr>        <dbl>
## 1 Kettle      35424.
## 2 Doritos     20926.
## 3 Pringles    16006.
## 4 Smiths      14928.
## 5 Infuzions    8749.
## 6 Twisties     7540.
## 7 Tostitos     7238.
## 8 Thins        7217.
## 9 Cobs         6145.
## 10 Red         4958.
## 11 Tyrrells     4801.
## 12 Grain        4201.
## 13 Cheezels     3318.
## 14 Natural      2130
## 15 Woolworths   1606.
## 16 Cheetos       899.
## 17 CCs          851.
## 18 French        429
## 19 Sunbites      391.
## 20 Burger       244.
```

```
create_bar_plot(
  data = revenue_by_brand_target_segment,
  x_col = BRAND_NAME,
  y_col = tot,
  title = "Total sales by brand - Mainstream young singles/couples",
  x_label = "Brand",
  y_label = "Total sales ($)"
)
```



We observe a clear dominance from the Kettle brand, which records more than \$35,000 in sales, while the runner-up, Doritos, sits at only \$20,925, and Pringles ranks third with \$16,000.

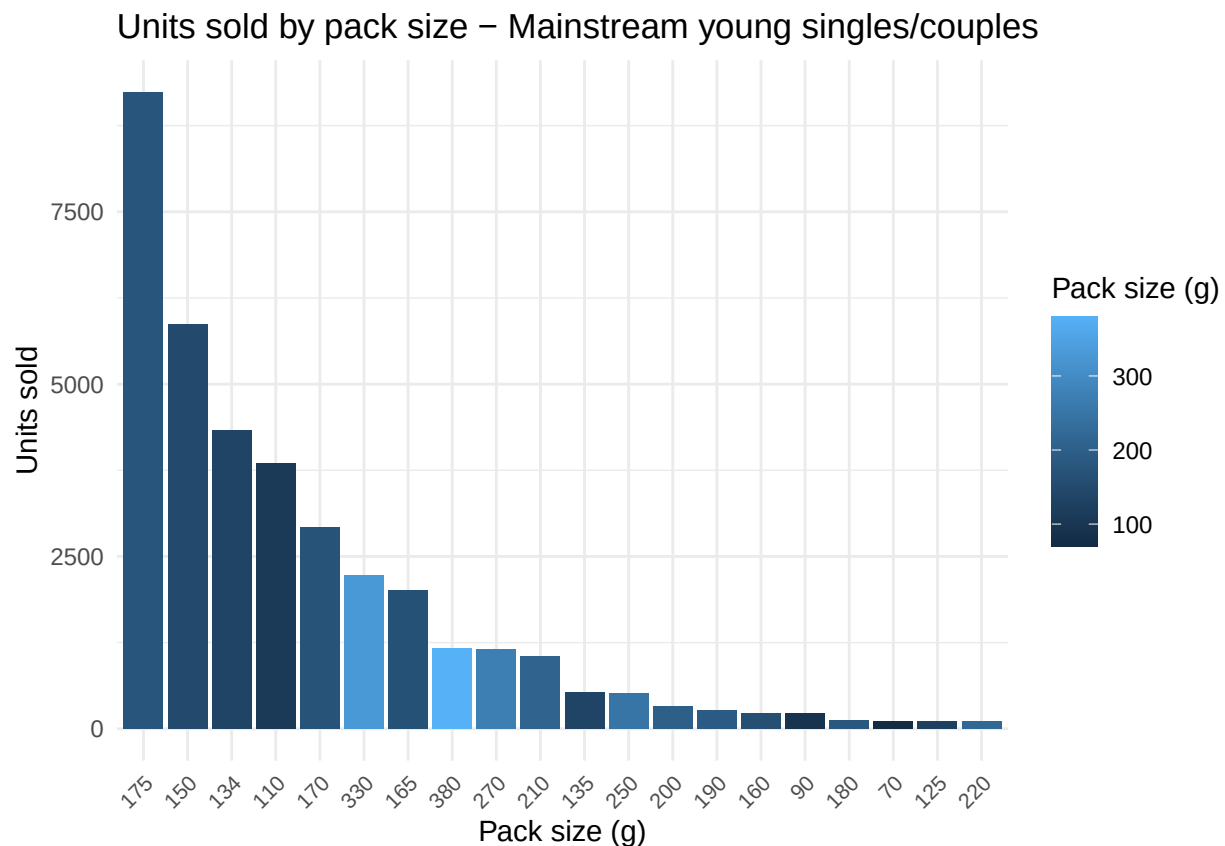
```
pack_size_target_segment <- chips |>
  filter(PREMIUM_CUSTOMER == 'Mainstream') |>
  filter(LIFESTAGE == 'YOUNG SINGLES/COUPLES') |>
  group_by(PACK_SIZE) |>
  summarise(units_sold = sum(PROD_QTY))

pack_size_target_segment |>
  arrange(desc(units_sold))
```

```
## # A tibble: 20 x 2
##   PACK_SIZE units_sold
##   <int>      <int>
## 1      175      9237
## 2      150      5863
## 3      134      4326
## 4      110      3850
## 5      170      2926
## 6      330      2220
## 7      165      2016
## 8      380      1165
## 9      270      1153
## 10     210      1055
## 11     135       535
```

```
## 12      250      520
## 13      200      325
## 14      190      271
## 15      160      232
## 16       90      230
## 17      180      130
## 18       70      110
## 19      125      109
## 20      220      106
```

```
create_bar_plot(
  data = pack_size_target_segment,
  x_col = PACK_SIZE,
  y_col = units_sold,
  title = "Units sold by pack size - Mainstream young singles/couples",
  x_label = "Pack size (g)",
  y_label = "Units sold"
)
```



We notice that the most popular pack size is 175g, with more than 9,000 units sold, while in second position we find 150g packs, with nearly 6,000 units sold, and rounding out the podium, 134g packs with 4,300 units.

```
brand_pct_target <- chips |>
  filter(PREMIUM_CUSTOMER == 'Mainstream', LIFESTAGE == 'YOUNG SINGLES/COUPLES') |>
  group_by(BRAND_NAME) |>
  summarise(units = sum(PROD_QTY)) |>
```

```

mutate(pct_target = units / sum(units) * 100)

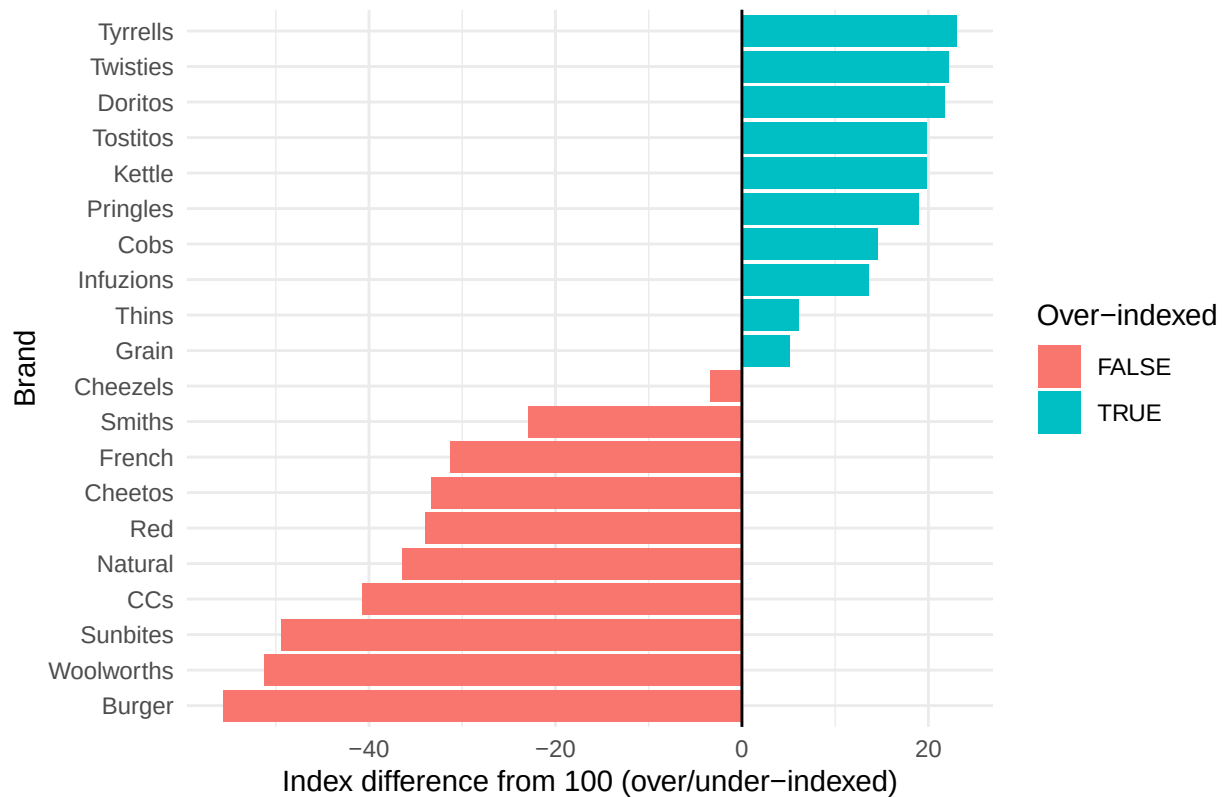
brand_pct_rest <- chips |>
  filter(PREMIUM_CUSTOMER != 'Mainstream' | LIFESTAGE != 'YOUNG SINGLES/COUPLES') |>
  group_by(BRAND_NAME) |>
  summarise(units = sum(PROD_QTY)) |>
  mutate(pct_rest = units / sum(units) * 100)

brand_affinity <- brand_pct_target |>
  select(BRAND_NAME, pct_target) |>
  left_join(brand_pct_rest |> select(BRAND_NAME, pct_rest), by = "BRAND_NAME") |>
  mutate(affinity_index = round(pct_target / pct_rest * 100, 1))

brand_affinity |>
  mutate(diff_from_avg = affinity_index - 100) |>
  ggplot(aes(x = reorder(BRAND_NAME, diff_from_avg), y = diff_from_avg, fill = diff_from_avg > 0)) +
  geom_col() +
  geom_hline(yintercept = 0) +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Brand affinity - Mainstream young singles/couples vs rest of population",
    x = "Brand",
    y = "Index difference from 100 (over/under-indexed)",
    fill = "Over-indexed"
  )

```

Brand affinity – Mainstream young singles/couples vs rest of population



We clearly observe a preference for certain brands within our target segment: brands such as Tyrrells, Twisties, and Doritos show sales volumes around 20% higher among our target group. Conversely, the Burger and Woolworths brands show a -40% difference.

Finally, regarding the brands previously identified as the most popular (Kettle, Doritos, and Pringles), we observe that they are also consumed more within this group, by an average of 20%, compared to the rest of the population.